Comparative Analysis of Depreciation Methods

Straight-Line vs. Diminishing Balance

Excel-Based Depreciation Calculator Project

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Based on the supplied project documentation and Excel calculator template.

1. Introduction

Depreciation is the systematic allocation of the depreciable amount of a tangible asset over its useful life. This project compares Straight-Line and Diminishing Balance methods using the provided Excel calculator.

2. Objectives

• Explain both depreciation methods.
• Calculate annual and total depreciation.
• Build a 10-year diminishing-balance schedule.
• Compare expense patterns and closing book values.

3. Inputs and assumptions

Input	Value
Asset cost	■450,000
Additional asset cost	■50,000
Total asset price	■500,000
Scrap value	■50,000
Useful life	10 years

Diminishing-balance rate assumption: 1 – (Scrap Value ÷ Asset Price)^(1 ÷ Useful Life), so the calculated book value reaches the scrap value at the end of the stated life.

4. Straight-Line method

Annual depreciation = (Asset Price – Scrap Value) ÷ Useful Life.

Annual depreciation = (■500,000 – ■50,000) ÷ 10 = ■45,000. The annual depreciation rate relative to total asset price is 9.00%. Total depreciation is ■450,000; closing book value is ■50,000.

5. Diminishing Balance method

Annual depreciation = Opening book value × depreciation rate. Calculated rate: 20.57%. Expense declines each year as the opening book value decreases.

6. Ten-year diminishing-balance schedule

Year	Opening value	Depreciation	Closing value
1	■500,000	■102,836	■397,164
2	■397,164	■81,685	■315,479
3	■315,479	■64,885	■250,594
4	■250,594	■51,540	■199,054
5	■199,054	■40,940	■158,114
6	■158,114	■32,520	■125,594
7	■125,594	■25,831	■99,763
8	■99,763	■20,518	■79,245
9	■79,245	■16,298	■62,946
10	■62,946	■12,946	■50,000

7. Comparative analysis

Measure	Straight-Line	Diminishing Balance
Annual expense	Constant ■45,000	Declines yearly
Year 1 depreciation	■45,000	■102,836
Total depreciation	■450,000	■450,000
Closing book value	■50,000	■50,000

8. Insights and conclusion

Straight-Line is simple and predictable. Diminishing Balance recognises more depreciation in earlier years and may be appropriate for assets that lose value or usefulness faster. The selection should follow the nature of the asset, the organisation’s accounting policy and applicable reporting requirements. Replace sample inputs with verified asset information before using the calculator for actual accounting decisions.